

Regression Algorithms and Their Explanations

1. Linear Regression Family

- Simple Linear Regression

Models the relationship between a single independent variable (X) and a dependent variable (Y) using a straight line.

Equation: $Y = aX + b$

- Multiple Linear Regression

Extends simple linear regression to include multiple independent variables.

Equation: $Y = a_1X_1 + a_2X_2 + \dots + a_nX_n + b$

- Polynomial Regression

Models the relationship as an nth-degree polynomial.

Useful for non-linear data that still looks like a curve.

Equation: $Y = a_0 + a_1X + a_2X^2 + \dots + a_nX^n$

- Ridge Regression

A type of linear regression that adds L2 regularization (penalty on large coefficients) to prevent overfitting.

- Lasso Regression

Similar to Ridge but uses L1 regularization, which can shrink some coefficients to zero (feature selection).

- ElasticNet Regression

Combines L1 and L2 regularization.

Useful when there are multiple correlated features.

2. Tree-Based Regression

- Decision Tree Regression

Splits data into branches to predict continuous values.

Non-linear and non-parametric.

- Random Forest Regression

Builds multiple decision trees and averages their predictions.

Reduces overfitting and increases accuracy.

- Gradient Boosting Regression

Builds trees sequentially, where each new tree corrects errors from previous trees.

- XGBoost Regression

An optimized version of gradient boosting with regularization and speed improvements.

- LightGBM Regression

A faster gradient boosting technique optimized for large datasets.

- CatBoost Regression

Handles categorical data automatically and reduces the need for extensive data preprocessing.

3. Support Vector Regression (SVR)

- Tries to fit the best line within a margin of tolerance.

- Works well with both linear and non-linear data using kernel tricks.

4. Instance-Based Learning

- K-Nearest Neighbors (KNN) Regression

Predicts the value based on the average (or weighted average) of the K nearest data points.

5. Probabilistic Models

- Bayesian Regression

Uses probability distributions to estimate model parameters.

Provides uncertainty estimation in predictions.

6. Neural Network Regression

- ANN (Artificial Neural Networks) Regression

Uses layers of interconnected neurons to model complex, non-linear relationships between input and output.

7. Other Specialized Regressions

- Quantile Regression

Models different quantiles (percentiles) of the data rather than the mean.

- Robust Regression

Designed to be less sensitive to outliers compared to traditional linear regression.

- Principal Component Regression (PCR)

First applies PCA to reduce dimensionality, then uses linear regression on principal components.

- Partial Least Squares Regression (PLSR)

Similar to PCR but tries to find components that correlate most with the output variable.